ELG 3125 - Signal & Systems

Lecture 1 - Part 1

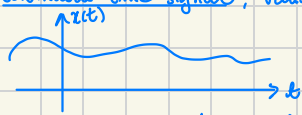
Introduction

Signal: data → function of time.

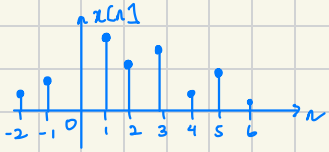
System: input → [system] → output, modelled using:
- Differential Equations (Difference Equations)
- Convolution integrals (sums)

Two types of signals:

- Continuous time signal, variable t → $x(t)$



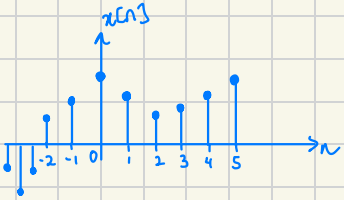
- Discrete time signal, variable n → $x[n]$

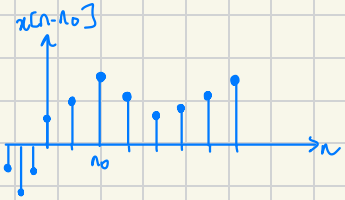


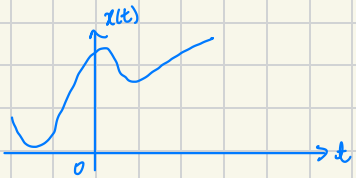
Transformations of the independent variable

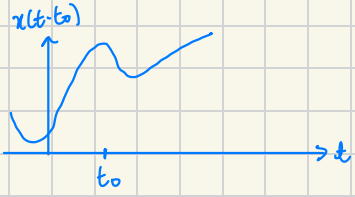
3 types of transformations:
- Reflection
- Scaling
- Shift

Shift

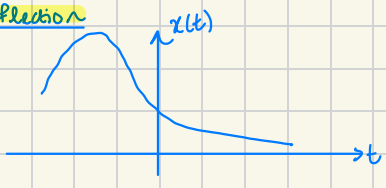


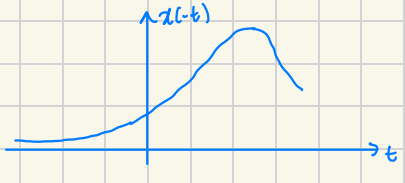






Reflection





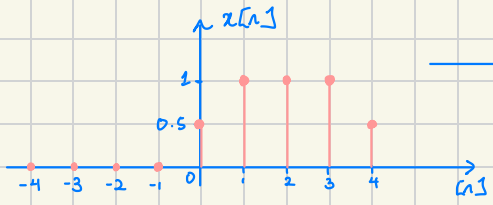
Scaling

For intuition: i) Multiplying shrinks
ii) Dividing enlarges

Order of operations

Shift → scale and reflect

Example: Given $x[n]$, plot $x[2n+1]$, $x[\frac{n}{2}+1]$



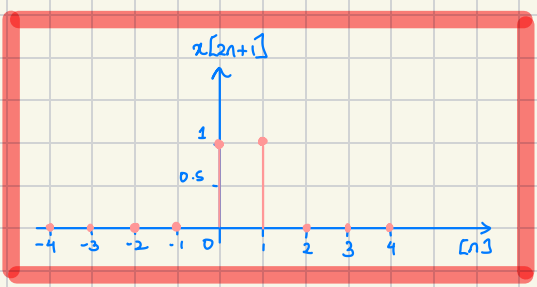
shift

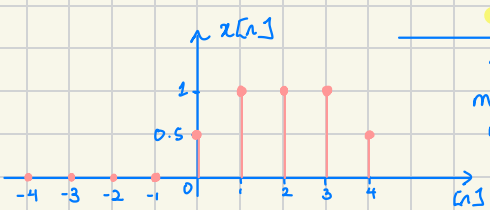
$x[n+1]$
move everything
left by
1



$\frac{-4}{2} = -2$ fraction, $\frac{-2}{2} = -1$ fraction, $\frac{0}{2} = 0$ fraction, $\frac{2}{2} = 1$ fraction, $\frac{4}{2}$ fraction

scaling process





shift

$x[n-1]$
move everything
right by
1



$-4(2) = -8$, $-3(2) = -6$, $-2(2) = -4$, $-1(2) = -2$, $0(2) = 0$, $1(2) = 2$, $2(2) = 4$, $3(2) = 6$, $4(2) = 8$, $5(2) = 10$

No fractions

scaling process

